CardioSim Single Cell Drug Simulation Result

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March 3, 2025

Simulation Parameter

Cellmodel: O'Hara-Rudy IKr-dynamic (2017)
Drug Name: quinidine
Cmax: 3237,6474,9711,12948
Ion Channel: 4 (INa, INaL, ICaL, IKr)

Simulation Protocol

Each drug was simulated by inducing 1000 beats at a 2000 ms cycle.
Biomarker: qNet, qInward, APD90, APD50, APDtri, CaD90, CaD50, CaDtri.
(1) qNet: Net charge carried by six major currents over a simulated beat (IKr, ICaL, INaL, Ito, IKs, IK1).

$$qNet = \int_0^{CL} (I_{NaL} + I_{CaL} + I_{Kr} + I_{K1} + I_{Ks} + I_{to})\, dt \tag{1}$$

(2) qInward: Inward currents carried by cations (such as sodium or calcium) into the cell during the depolarization phase of the cardiac action potential.
(3) APD90: Action Potential Duration at 90% repolarization. time it takes for the action potential of a cardiac cell to repolarize to 90% of its maximum
(4) APD50: APD at 50% repolarization. time it takes for the action potential of a cardiac cell to repolarize to 50% of its maximum
(5) APDtri: Triangulation of the APD. Difference between APD90 and APD50.
(6) CaD90: Calcium Duration at 90% recovery. Time it takes for the concentration of calcium ions within a heart cell to recover to 90% of its peak level.

(7) CaD50: Calcium Duration at 50% recovery. Time it takes for the concentration of calcium ions within a heart cell to recover to 50% of its peak level
(8) CaDtri: Triangulation of the CaD. Difference between CaD90 and CaD50.

Method

Calculate biomarkers by simulating one cycle after assigning the state values of the ventricular cell membrane protein gates derived through in silico simulation as the initial conditions of the model.

Result

Time-series Result

The following results will show the time-series result of the action potential, rates of action potentials, intracellular calcium, and ionic currents involved in the drugs.

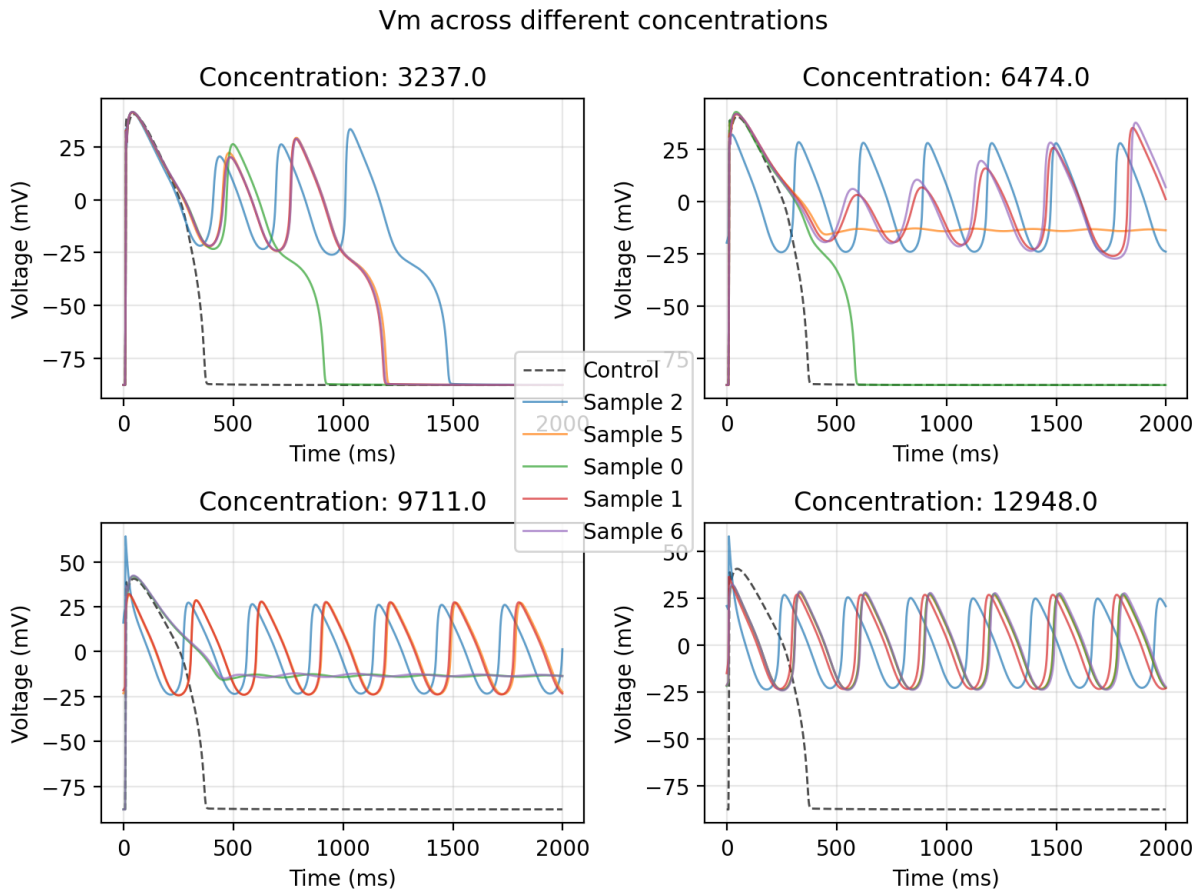

Figure 1: Action potential result in various concentrations.

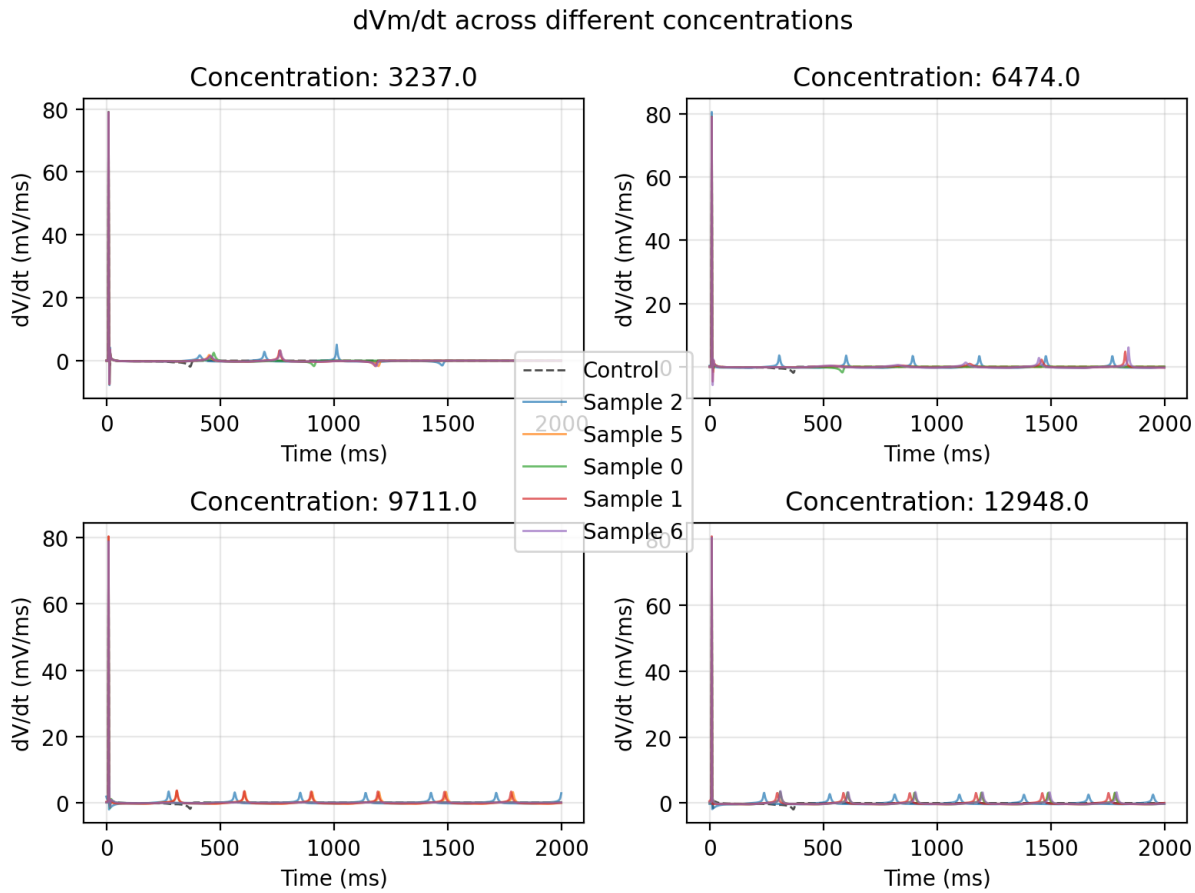

Figure 2: Rates of action potential result in various concentrations.

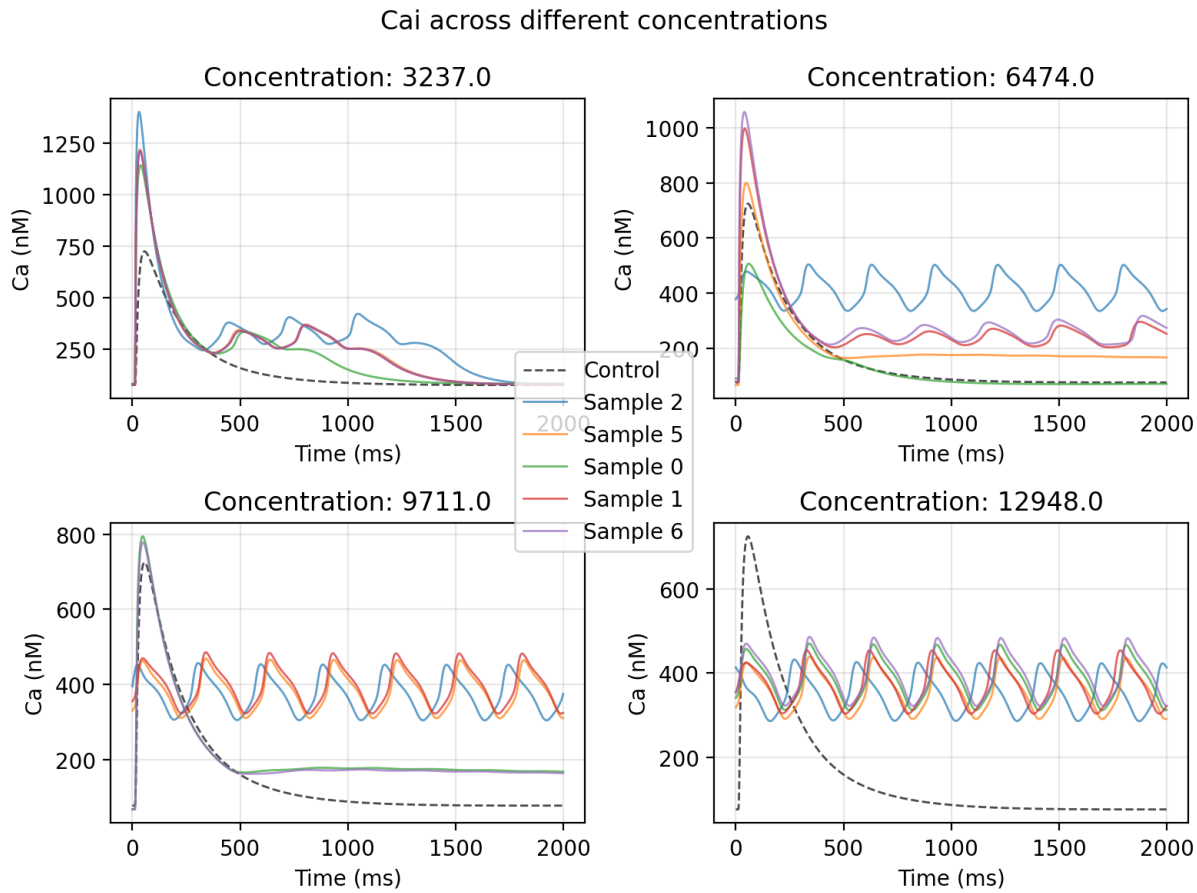

Figure 3: Intracellular calcium concentration result in various concentrations.

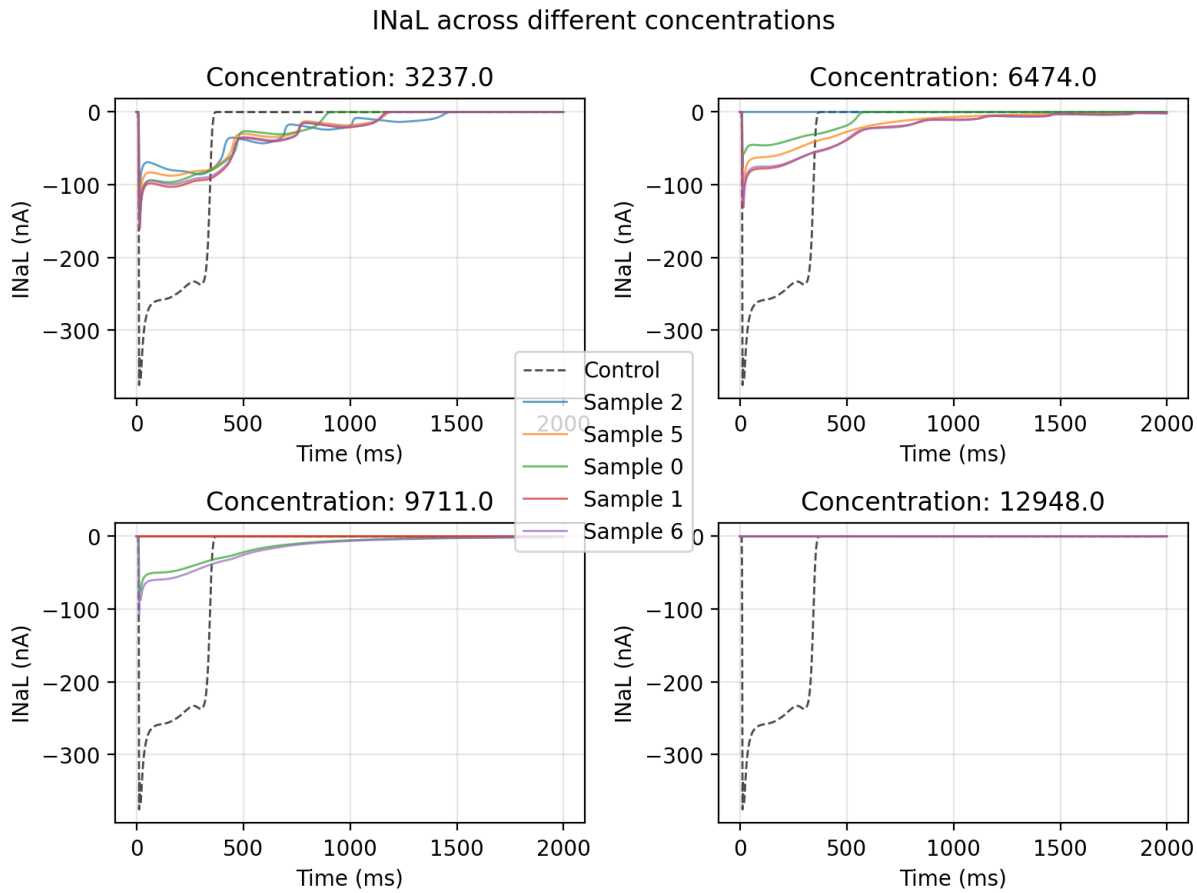

Figure 4: L-type sodium current result in various concentrations.

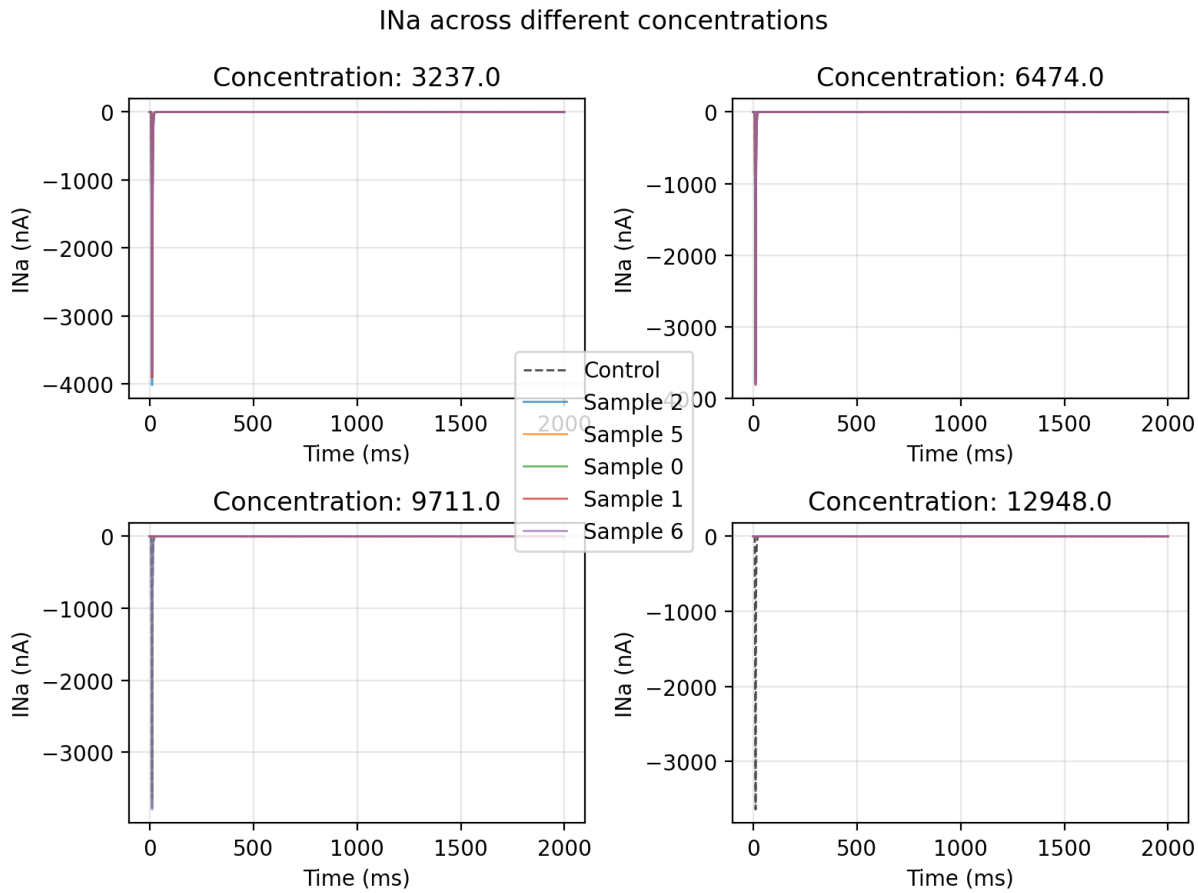

Figure 5: Sodium current result in various concentrations.

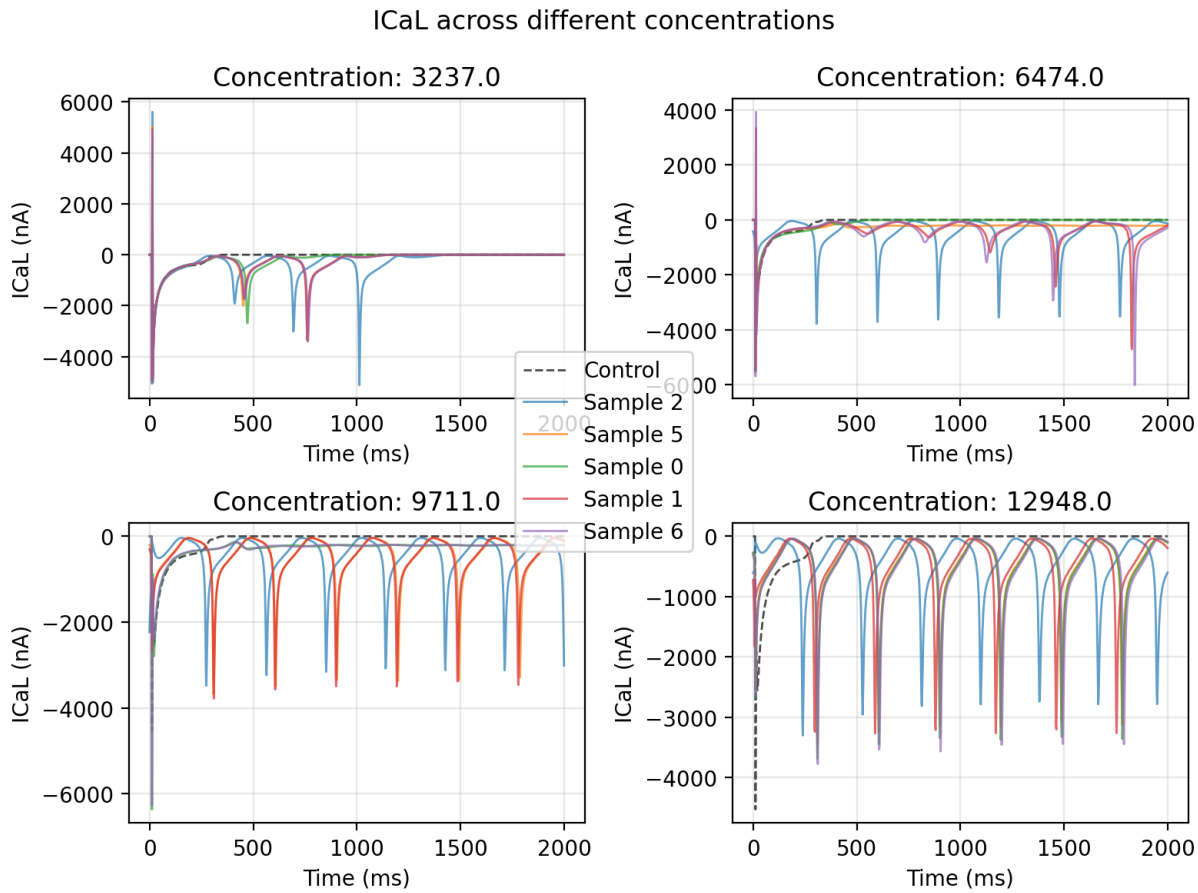

Figure 6: L-type calcium current result in various concentrations.

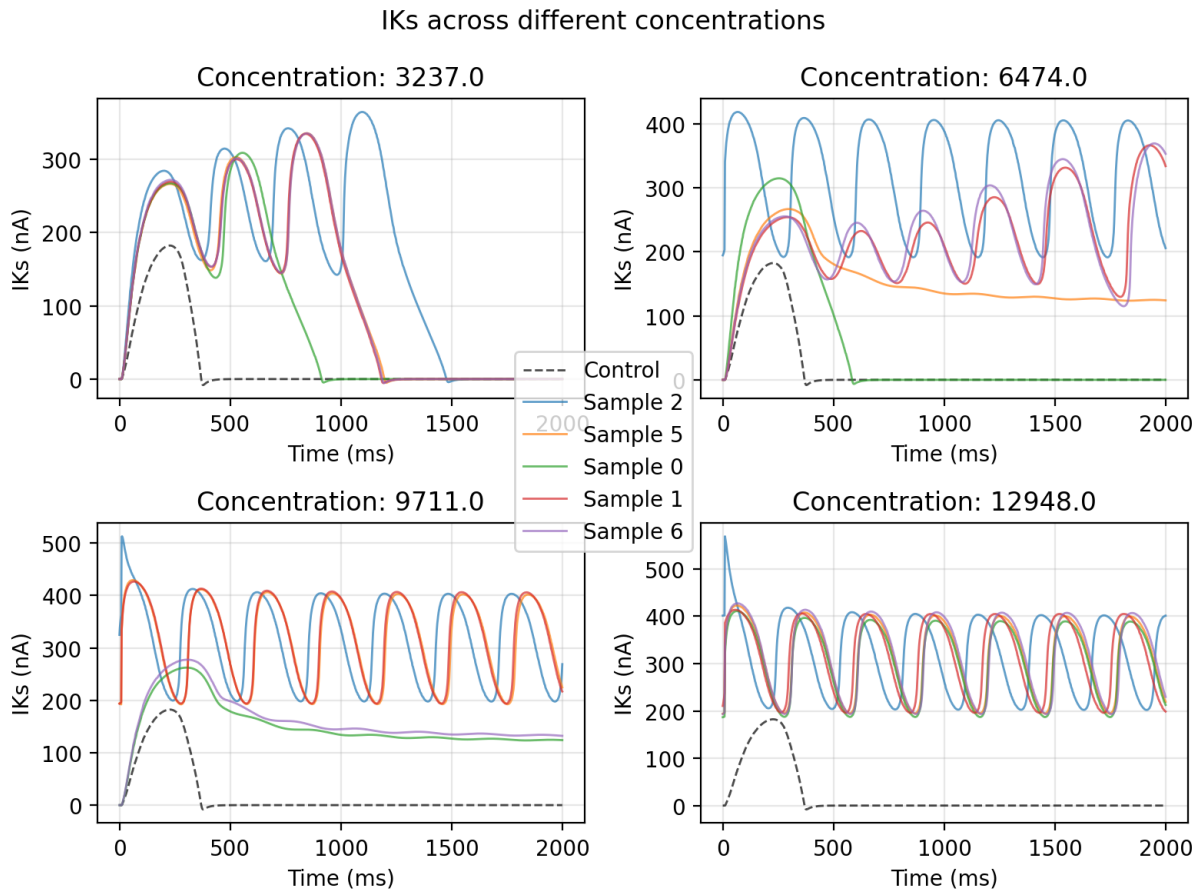

Figure 7: Slow potassium current result in various concentrations.

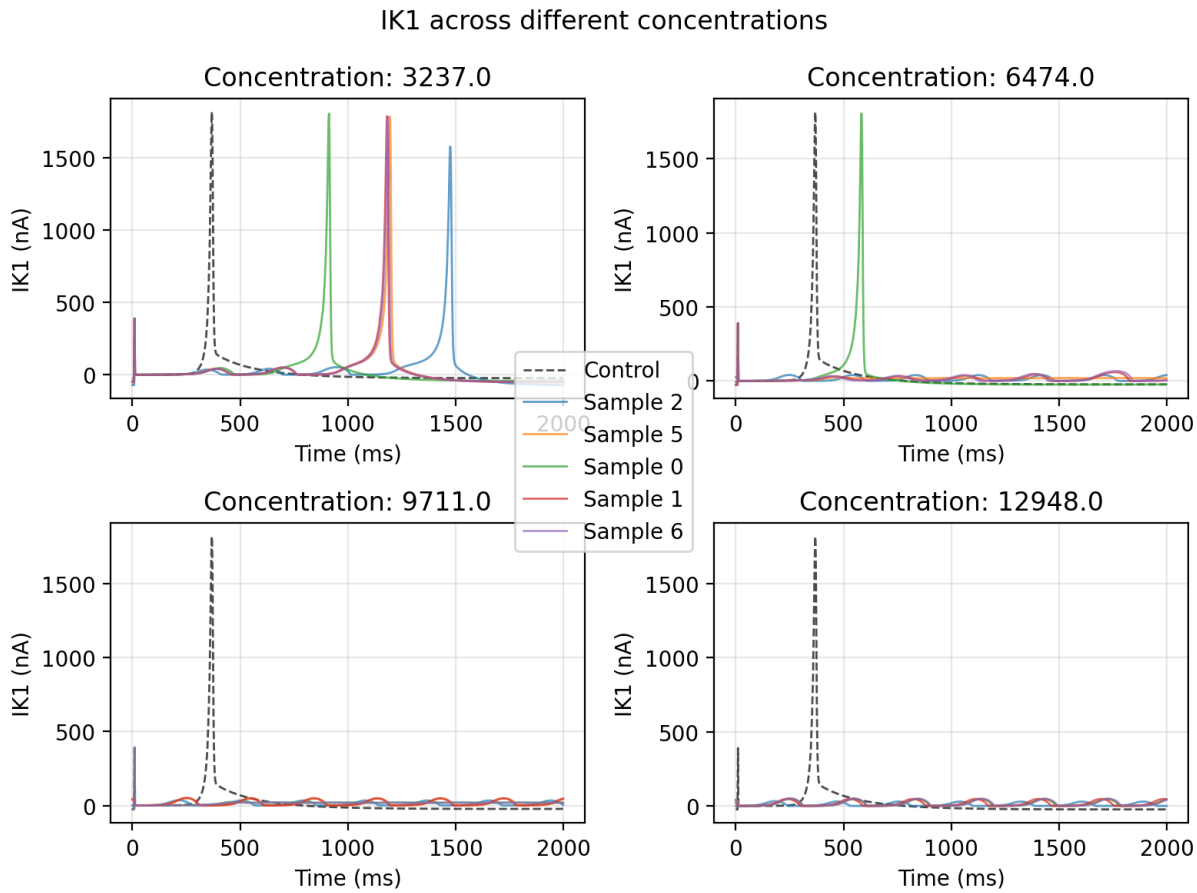

Figure 8: Inward rectifier current result in various concentrations.

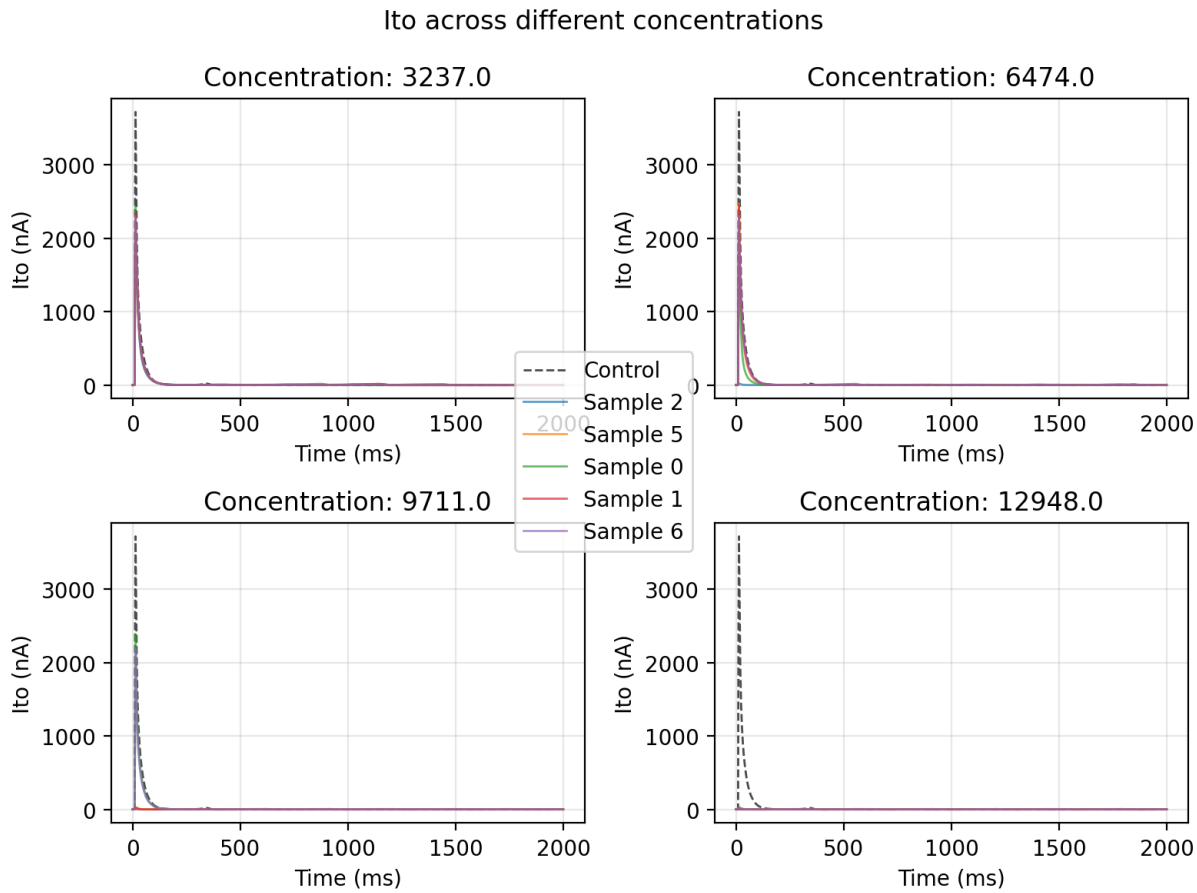

Figure 9: Transient outward current result in various concentrations.

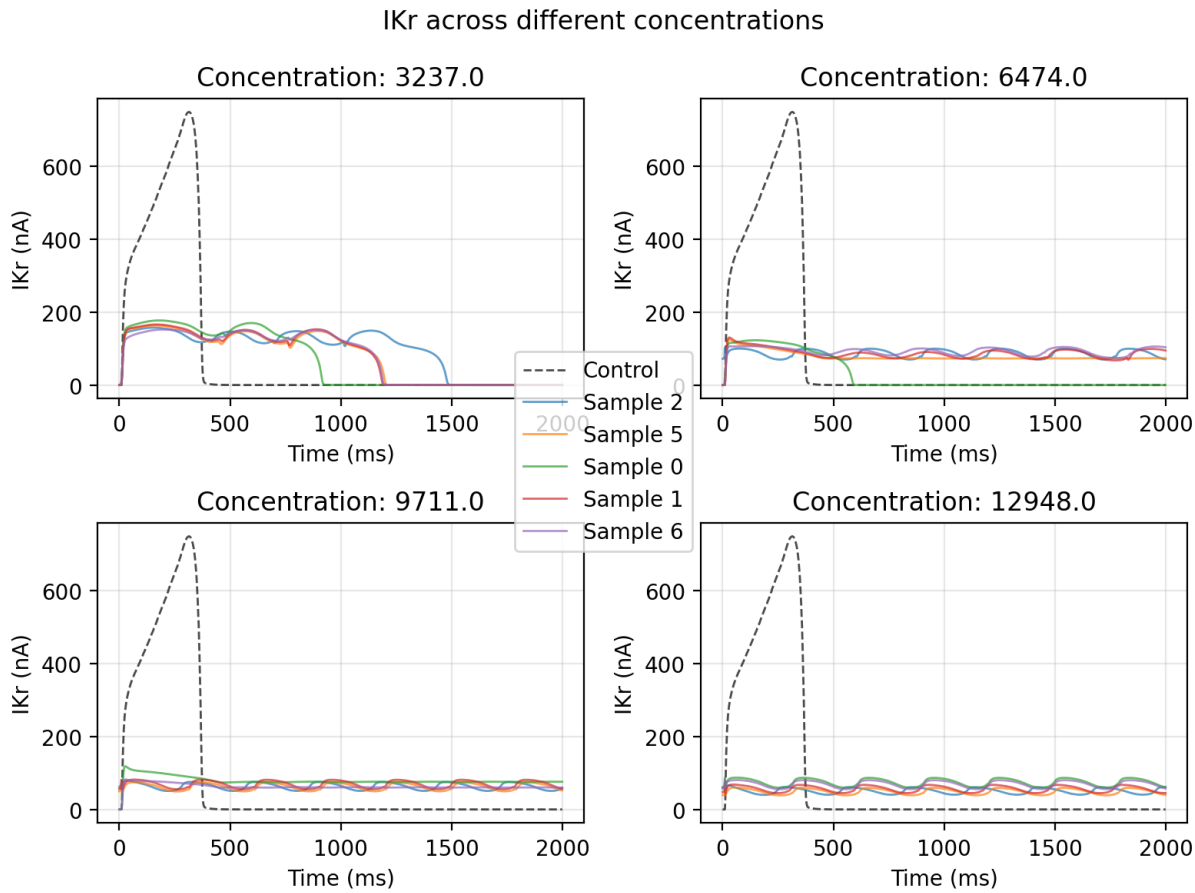

Figure 10: hERG current result in various concentrations.

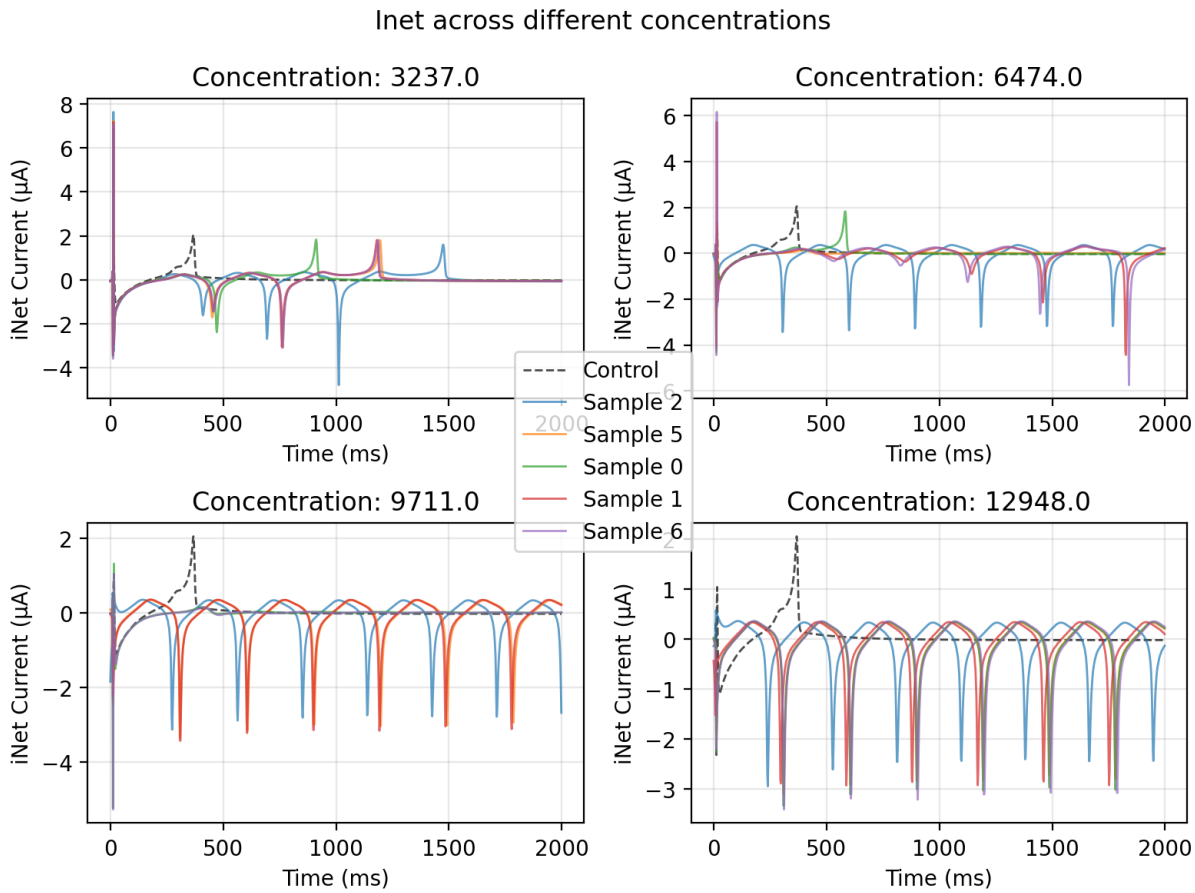

Figure 11: Cummulative current result (iNet) in various concentrations.

Feature Distribution

The biomarker for cardiotoxicity assessment proposed by the US FDA Leading Study Group is qNet, and the following figures show the distribution of qNet and other features.

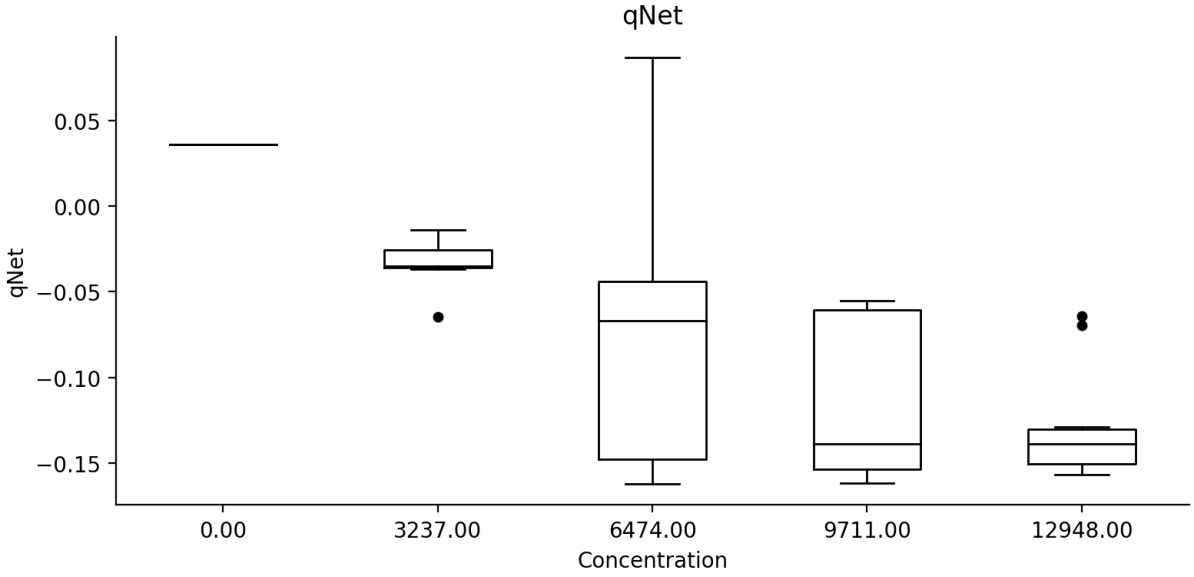

Figure 12: qNet value distributions among several concentrations.

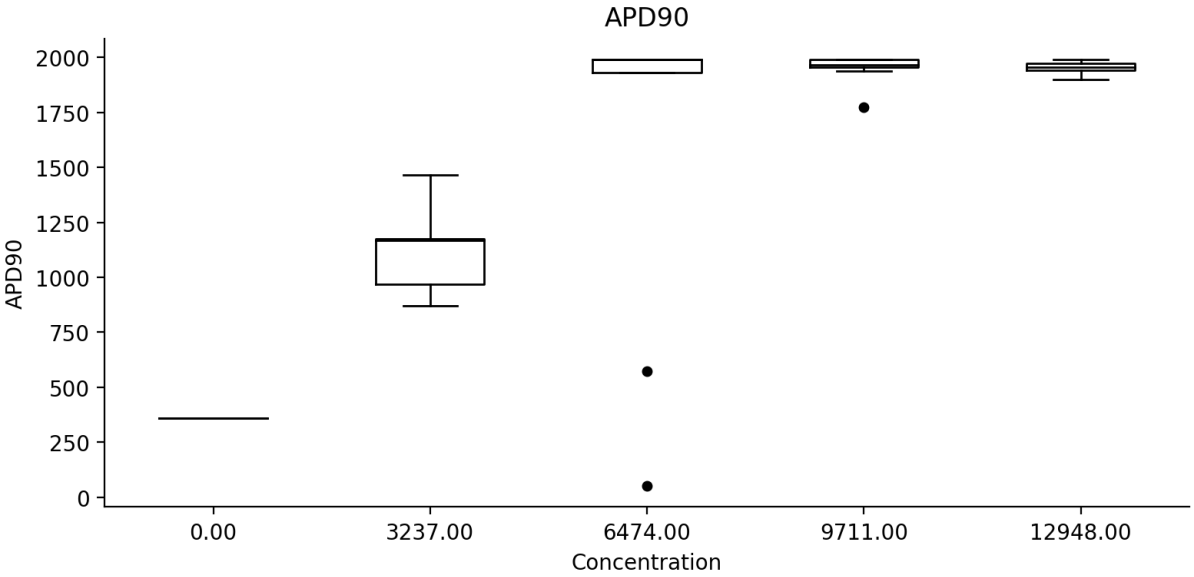

Figure 13: APD90 value distributions among several concentrations.

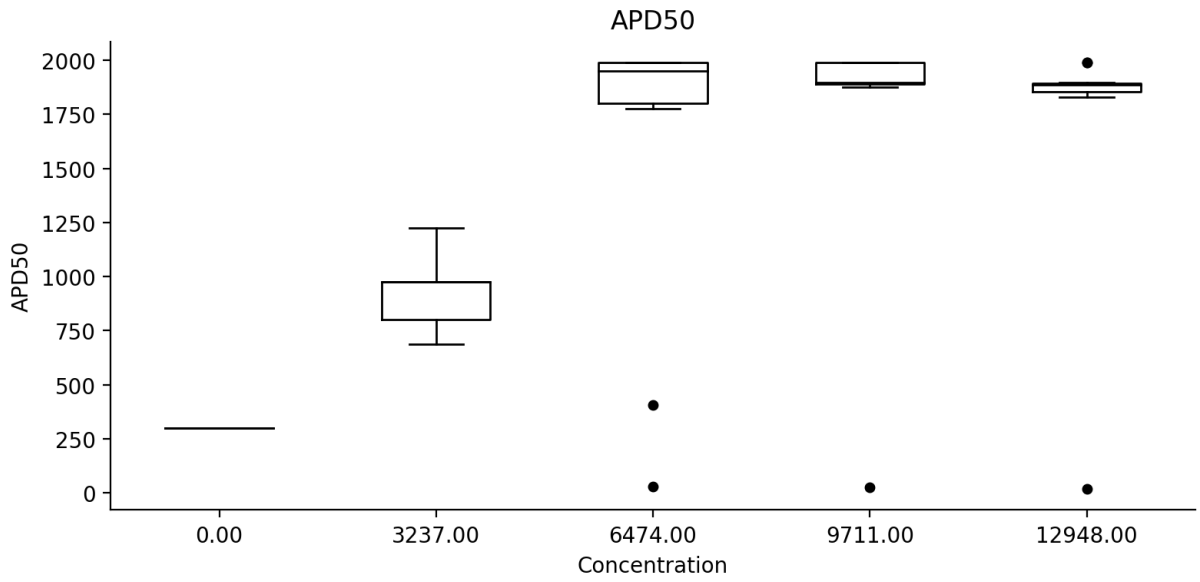

Figure 14: APD50 value distributions among several concentrations.

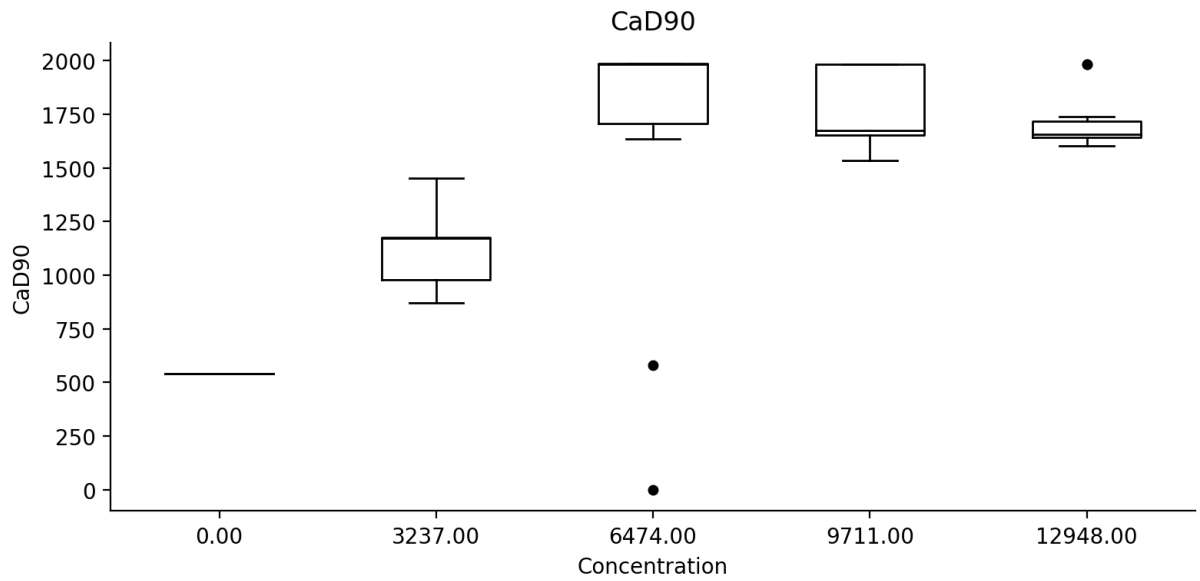

Figure 15: CaD90 value distributions among several concentrations.

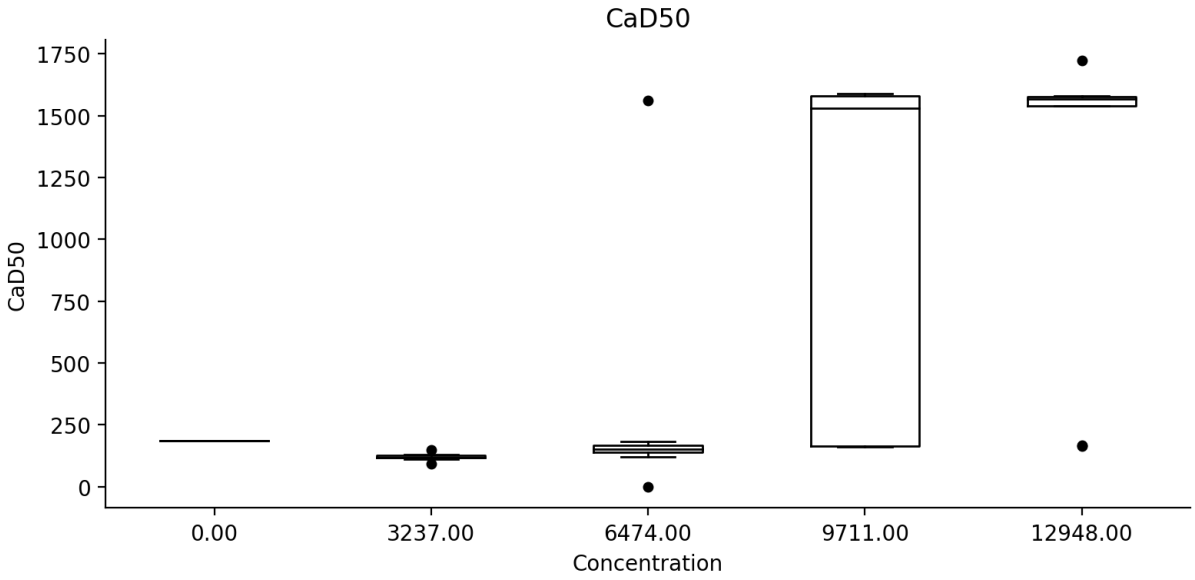

Figure 16: CaD50 value distributions among several concentrations.

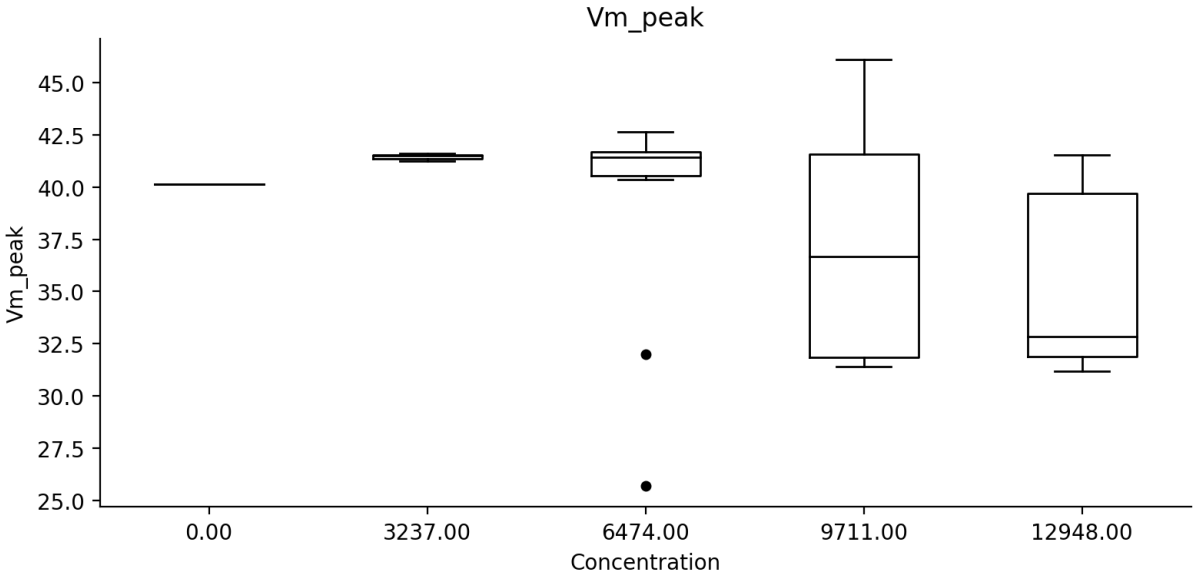

Figure 17: Peak action potential value distributions among several concentrations.

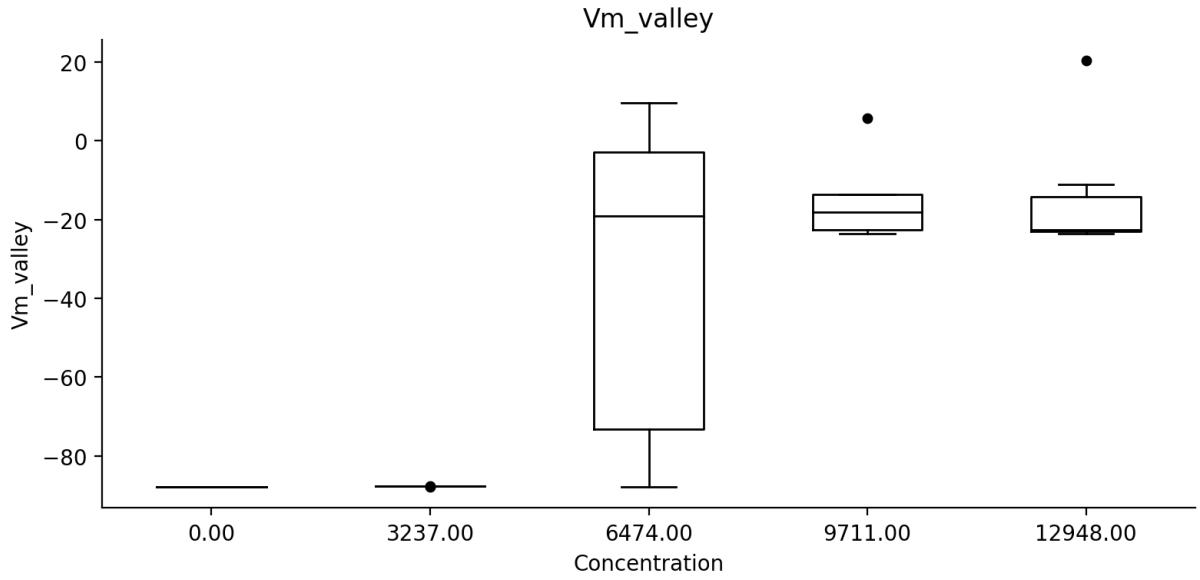

Figure 18: Lowest action potential value distributions among several concentrations.

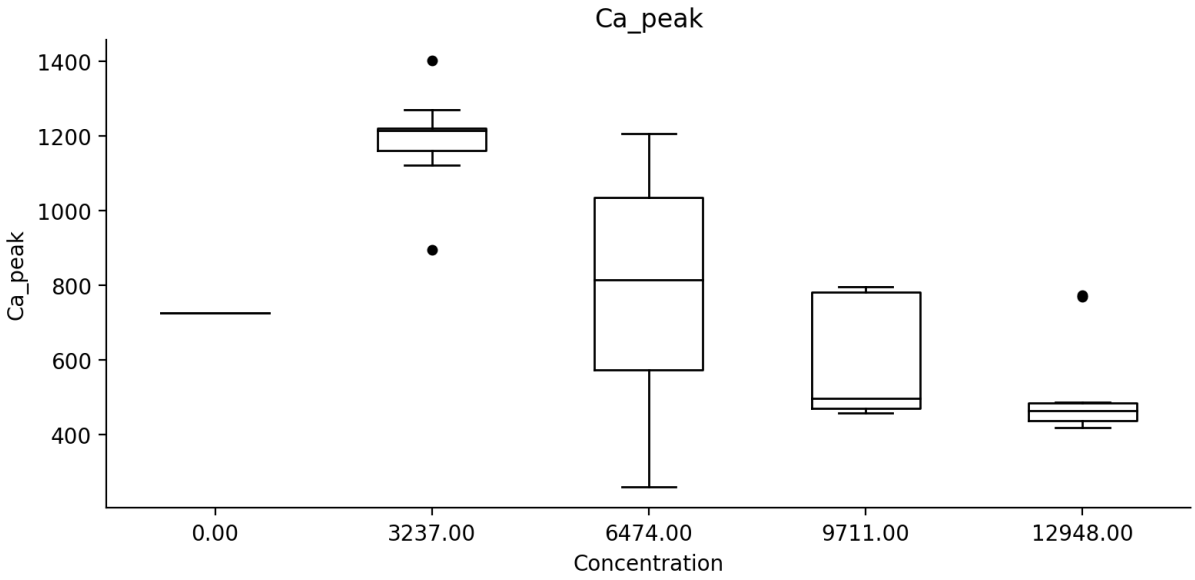

Figure 19: Peak calcium concentration value distributions among several concentrations.

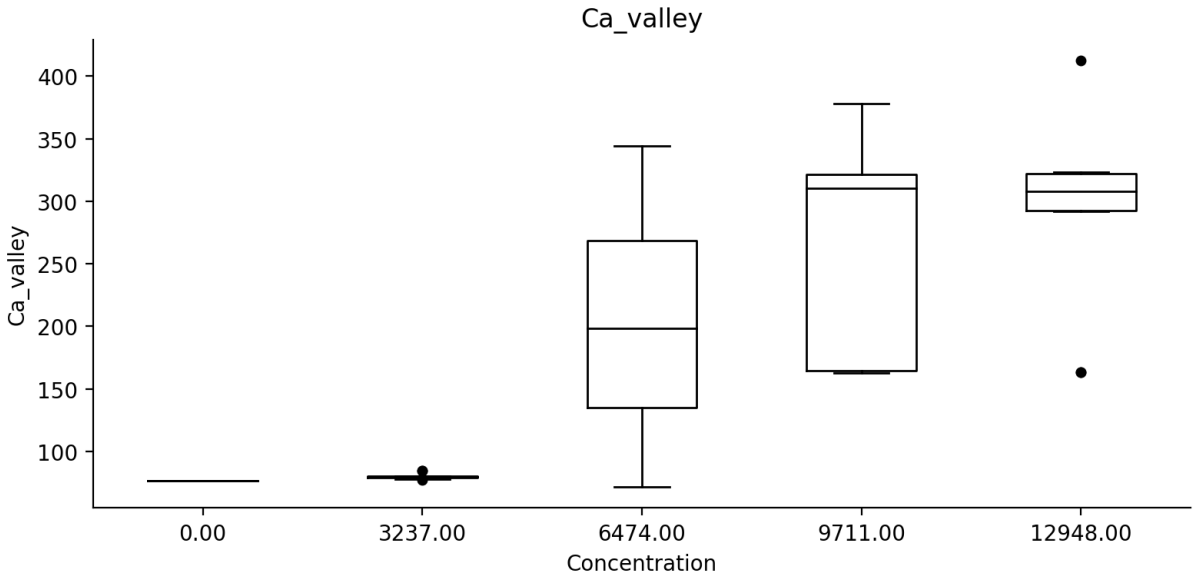

Figure 20: Lowest calcium concentration value distributions among several concentrations.